

**Time:** 5 minutes**Outcome:** Apply Shutdown & Short-Run Loss Minimization.**Difficulty:** ●●○ Intermediate

Shutdown & Short-Run Loss Minimization



THE CORE IDEA

A firm should operate in the short run when price covers average variable cost, even if price is below average total cost. A lump-sum tax acts like fixed cost. AVC represents avoidable cost in the short run. Short-run loss minimization compares the incremental benefit and avoidable cost of operating.



HOW TO RECOGNIZE IT

- Find the profit-maximizing or loss-minimizing output where marginal revenue equals rising marginal cost.
- At that output, compare price with average variable cost.
- Operate in the short run when price covers average variable cost; shut down when price falls below it.
- A firm may operate at a loss when it covers variable cost and contributes something toward fixed cost.

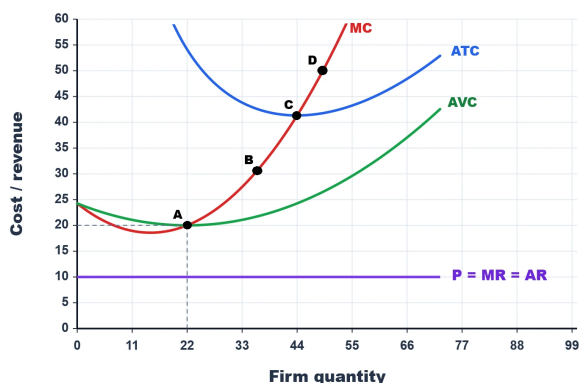


WATCH OUT

A common mistake is to misapply competitive firm rule. Short-run loss minimization compares the incremental benefit and avoidable cost of operating. A fixed-cost change affects profit and ATC but not MC or AVC, so it does not change the marginal output choice or shutdown threshold.



WORKED EXAMPLE: SHUT DOWN OR OPERATE?



The horizontal price line is \$10, below minimum AVC of \$20 at point A. At every positive output, revenue per unit would be less than avoidable variable cost. The firm minimizes its short-run loss by shutting down and paying only fixed cost.



CHECK YOURSELF

P is below minimum AVC. What should the firm do?

**READY?**

Return to the game and master this concept.